Sparse Recovery Above the Phase Transition: A Learned Operator-Aware Approach

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***Abstract*—We characterize the sparsity regime where learned sparse recovery has genuine headroom over classical methods. Below the ℓ_1 phase transition (k=25 in n=256, m=128), CoSaMP [2] achieves exact recovery on both partial Fourier and Gaussian operators, while LISTA [8] trained on Fourier and tested zero-shot on Gaussian plateaus at NRMSE 0.33. At k=55 (just above the Donoho–Tanner weak threshold [6]), polynomial-time greedy methods fail (CoSaMP=0.55, OMP=0.90 [3] NRMSE) while oracle-support least squares remains exact. A coordinate-wise learned support detector with operator-aware features beats coherence-only baselines (25–34% relative NRMSE) but does not match CoSaMP. The over-sparse regime is the open setting where learned recovery has a defensible role, with two structural advantages — adaptive cardinality and amortized signal-prior inference — independent of regime. On Gaussian compressible signals the same detector, trained on mixed-tail-amplitude signals, beats CoSaMP by 0.074–0.097 NRMSE (8–10% relative) at off-support tail amplitudes 0.3–0.4 (where CoSaMP's NRMSE exceeds 1); the gap is structural, holding in 15 of 15 runs across 5 operator draws $\times$ 3 initializations. The broader aim is a learning approach that beats CoSaMP across all sparsity regimes — the compressibility win is the first concrete instance.**

I. Below the Transition: Classical Sparse Recovery Wins

We consider the sparse linear inverse problem $y = Ax + \varepsilon$ [1] with $A \in \mathbb{R}^{m\times n}$ fixed at deployment and $\|x\|_0 = k$. Classical baselines (CoSaMP [2], OMP [3], AMP [4]) and learned-recovery work (LISTA [8] and variants [9]–[11]) are typically benchmarked well below the ℓ_1 phase transition [6]. At n=256, m=128, k=25 (noiseless), CoSaMP achieves NRMSE ≈ 0 on both partial Fourier (μ=0.75) and Gaussian (μ=0.37) matrices. By contrast, LISTA trained on Fourier and tested zero-shot on Gaussian achieves NRMSE 0.33; the best parameter-free support-then-amplitude baseline achieves 0.26. The learning "advantage" here is illusory: CoSaMP [2] dominates by an order of magnitude in this strict-sparse regime. (The picture changes once signals become compressible rather than exactly k-sparse: see Section V.)

II. Above the Transition: Learning Approach Has the Upper Hand

The landscape inverts above the phase transition. Table I reports NRMSE on Gaussian at k=55 (k/n=0.21, k/m=0.43, just above the Donoho–Tanner weak threshold $\rho_W\approx$0.4): signals remain sparse, but classical guarantees no longer hold. All polynomial-time greedy methods fail by 0.55–0.90 NRMSE while oracle support recovery remains exact: the recovery problem is well-posed but combinatorial — the regime in which learning approaches have headroom (Section V).

TABLE I
Gaussian operator (n=256, m=128, 500 test signals). NRMSE at k=55, above the ℓ_1 phase transition.

Method	NRMSE
Naive top-k on $\lvert A^\top y\rvert$ + LS	0.79
OMP	0.90
CoSaMP	**0.55**
Learned detector (ours)	0.59
Oracle support + LS	**0.00**

III. Operator-Aware Support Detector

For each coordinate j we build a feature vector $\phi_j = \left[|x_j^{(T)}|, A_{:,j}^\top(y-Ax^{(T)}), A_{:,j}^\top y, (A^\top A)_{jj}, \mu_j\right]$ with $x^{(T)}$ a T=30-step ISTA [5] iterate and $\mu_j = \max_{i\neq j} |A_{:,i}^\top A_{:,j}|$. The residual correlation encodes whether j can still explain the measurement residual; the Gram diagonal and coherence describe local identifiability under the deployment operator. A coordinate-wise MLP (4,609 params) maps each ϕ_j to a support probability; the top-k scoring indices form the predicted support, with amplitudes recovered by support-restricted least squares.

Trained on 4,000 k=55-sparse signals and evaluated on 500 held-out signals, the detector achieves NRMSE 0.59 vs 0.79 for naive top-k and 0.90 for OMP; it does not match CoSaMP's 0.55. Test IoU plateaus at 0.51 by epoch 10 — an information ceiling, not an optimization one.

IV. Discussion

The CoSaMP gap is structural: CoSaMP performs implicit set-level reasoning (merge $2k$, refit, prune), while a coordinate-wise detector scores each index from its own feature vector with no view of competing candidates. The IoU plateau at 0.51 pins this as a representational ceiling, not an optimization one. Two further structural advantages of learned detectors hold *independently* of the recovery regime. First, CoSaMP requires k as input, while a learned threshold-based detector can estimate cardinality from data — real signals are not exactly k-sparse and the true k is rarely known. Fig. 1 (right column) shows the qualitative trend; Section V turns it into a quantitative win. Second, classical methods are per-instance

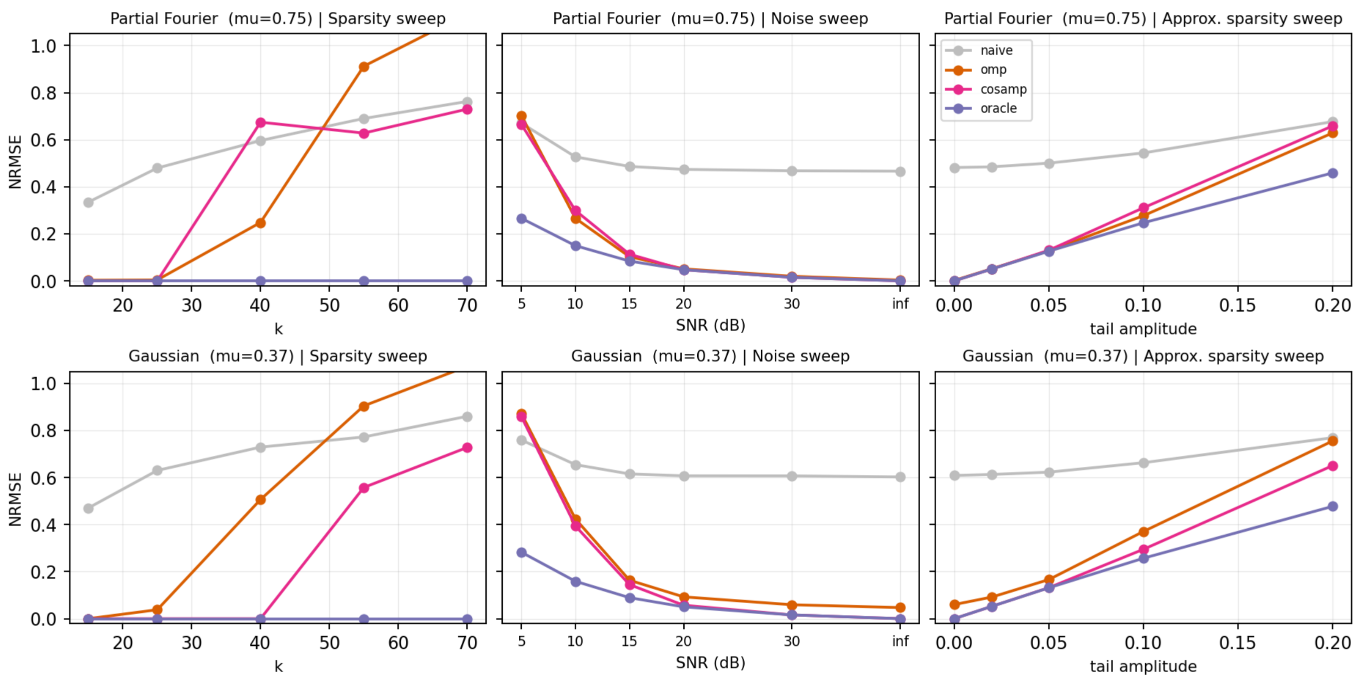

Fig. 1. Stress tests on partial Fourier (μ=0.75, top) and Gaussian (μ=0.37, bottom) operators at n=256, m=128. **Left:** sparsity sweep—CoSaMP and OMP achieve exact recovery below the phase transition but degrade sharply past k≈40, while oracle support recovery remains exact throughout. **Center:** noise sweep at k=25—all methods track the oracle closely above 15 dB. **Right:** approximate-sparsity sweep—as off-support tail amplitude grows, oracle performance itself degrades and the greedy–naive gap narrows, exposing CoSaMP's reliance on exact k-sparsity.

and oblivious to signal priors, while learning amortizes prior structure across the training distribution — a gain that should compound for signals with exploitable structure (block-sparse, transform-domain compressible).

V. Beyond CoSaMP: An Empirical Win Under Compressibility

The structural argument — that CoSaMP's strict k-sparsity assumption becomes the bottleneck once signals are compressible rather than exactly sparse — is backed by a direct empirical win. Training the same coordinate-wise detector on Gaussian compressible signals [7] with random off-support tail amplitude $\sim$Unif$[0.1, 0.4]$ (k=25, m=128, labels = top-k of $|x|$) and evaluating at fixed tail amplitudes (500 test signals each):

tail amplitude	0.1	0.2	0.3	0.4
CoSaMP	**0.28**	0.63	0.88	1.01
Learned (ours)	0.37	**0.62**	**0.80**	**0.91**
Δ (gain)	−0.08	+0.01	**+0.07**	**+0.10**

NRMSE; bold marks the lower (better) value per column. Δ= CoSaMP − Learned, so positive favors the learned detector.

Each cell reports the mean across 5 operator seeds $\times$ 3 model initializations (15 runs); the wins at tail 0.3 and 0.4 hold in 15 *of* 15 *runs* with cross-run std $\leq$ 0.003 NRMSE (paired-bootstrap 95% CI: $[+0.073, +0.076]$ at tail 0.3, $[+0.096, +0.099]$ at tail 0.4). The result is structural, not a single-shot artifact: CoSaMP wins at low tail (its strict-k assumption nearly holds) but collapses past tail 0.2; at 0.4 its NRMSE exceeds 1, worse than predicting zero. The learned detector crosses CoSaMP at tail 0.2 and pulls 8–10% ahead at 0.3–0.4, the only method that stays below the 1.0 floor. The mechanism is exactly what the structural argument predicted: top-k of $|x|$ now includes the largest tail entries, which CoSaMP — locked to the original spike support — cannot select. The low-tail loss is also structural: widening training to Unif$[0, 0.4]$ does not close it, indicating that per-coordinate features cannot match CoSaMP's set-level merge-and-refit at the strict-sparse end. Note also that on strict-sparse k=55 Gaussian the detector already matches CoSaMP's support IoU (0.51 vs 0.53), so the residual NRMSE gap there is amplitude- not support-bound — a clue Section VI follows up.

VI. Preliminary: Attention as a Set-Level Mechanism

The IoU plateau at 0.51 motivated a single-block remedy: replace the per-coordinate MLP with a small transformer encoder [12] (n=256 coordinate tokens, embedding dim 24, 4 heads, 2 layers, ∼10k params), so each index can see its competitors. We inject operator awareness through an additive Gram-matrix attention bias: for each head h, the attention logits gain $\alpha_h \cdot |G_{ij}|$ before the softmax, where G=$A^\top A$ with the diagonal zeroed and α_h is a learnable scalar initialized at 0. We call this GAST — the Gram-Aware Support Transformer. Two preliminary single-seed findings, paired against the MLP on the same operator and training data:

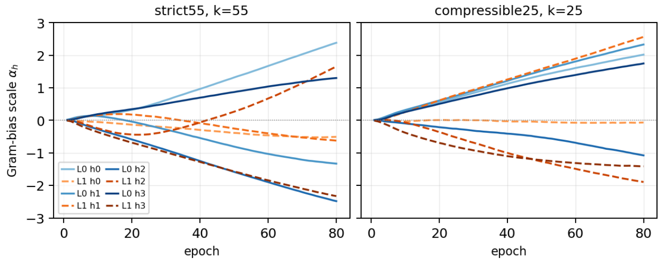

Fig. 2. Per-head, per-layer Gram-bias scaling α_h over 80 training epochs. Solid: layer 0; dashed: layer 1. Both regimes drive α to magnitudes of ± 2.5 with a mix of positive (attend *more* to coherent neighbors) and negative (attend *away*) heads. The mechanism engaged equivalently in both regimes, yet only compressible translates that engagement into a test-time win.

(i) Compressibility extension. On compressible $k{=}25$ Gaussian (Unif$[0.1, 0.4]$ training), GAST extends the MLP's CoSaMP win in every cell of the §V table: at tail$=$ 0.4, GAST beats CoSaMP by 0.106 NRMSE versus MLP's 0.097; at tail$=$ 0.3, $+0.080$ versus $+0.075$. Modest but consistently in the right direction.

(ii) Strict-sparse no-improvement. On strict $k{=}55$ Gaussian, GAST and MLP land within 0.001 NRMSE and 0.001 IoU of each other — both plateau, both lose to CoSaMP by $\sim$0.034. The 0.51 IoU ceiling holds.

The negative result is informative because α *did* engage (Fig. 2): magnitudes reach ± 2.5 in both regimes, with positive heads (focus on competing candidates) and negative heads (suppress confusable neighbors); trajectories are essentially indistinguishable across regimes, yet only compressible yields a test gain. The IoU plateau is therefore *not* a per-coord-vs-set-level distinction — set-level reasoning was added with maximum expressiveness, α used it, and test performance still did not move. The remaining bottleneck on strict-sparse is more likely either an information ceiling in the 5-feature interface, or the absence of iterative algorithmic structure (CoSaMP's merge-refit-prune loop runs against the *current* residual; neither MLP nor a single-block GAST does).

VII. Outlook

The preliminary GAST result reframes the next-architecture question: single-block attention is necessary but not sufficient at strict-sparse $k{=}55$; the natural escalation is iterative algorithmic structure — a learnable analogue of CoSaMP's merge-refit-prune loop with attention modules at each step. Two experiments would convert the regime-independent advantages from claim to evidence: *(i)* training on block- or cluster-sparse priors to measure the amortization gain; and *(ii)* replacing top-k with a learned threshold to test adaptive cardinality directly. Together — iterative attention, learned threshold, structured priors — these directions target all three of CoSaMP's structural limitations (set-level reasoning, prior-blindness, oracle-k dependence), aiming at a learned operator-aware framework that uniformly outperforms CoSaMP regardless of sparsity regime.

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